Text-to-Image Generation on MS COCO with Stable Diffusion 1.5: Milestone 1

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Dataset

We built a 2,000-pair image-caption subset of MS COCO 2017, balanced across all 80 object classes (25 images per class, seed 42). Captions were lowercased, cleaned and de-duplicated, and every image was resized to 512x512, the native resolution of Stable Diffusion 1.5. The subset is stored in `data/subset.csv` (image_id, file_name, processed_path, caption, category, supercategory) and can be rebuilt with `python src/data.py`.

Model

We use Stable Diffusion 1.5 through Hugging Face Diffusers, with PyTorch and Transformers. Its text encoder is CLIP ViT-L/14, which maps each caption to a 77 x 768 embedding that conditions the denoising U-Net through cross-attention; a VAE decodes the final latent into a 512x512 image. Milestone 1 builds the caption-to-embedding pipeline on its own and runs the full pipeline unmodified as our baseline.

Roles

Part 1, Data (Sai Tanoj Salehundam). Selected the balanced 2,000-pair COCO subset, cleaned captions, resized images to 512x512 and wrote `src/data.py` and `data/subset.csv`.

Part 2, Embeddings + Environment (Raymond Wang). Set up the environment and `requirements.txt`, and built `src/embed.py` and `notebooks/baseline_embedding.ipynb`, which load the CLIP ViT-L/14 text encoder, embed all 2,000 captions (77 x 768 each) and check that similar captions produce similar embeddings.

Part 3, Tests (Rajiv Praveen). Wrote `notebooks/tests.ipynb`, validating the subset and embeddings and producing a sample image-caption grid (see Sample testings).

Part 4, Generation + Report (Tanatswa Manzini). Wrote `src/generate.py`, which runs baseline SD 1.5 on 5 subset captions (one per supercategory) at 512x512, 25 steps, guidance 7.5 and fixed seeds 42 to 46, saving images, a `generations.csv` log and a comparison grid to `outputs/`. Assembled this report.

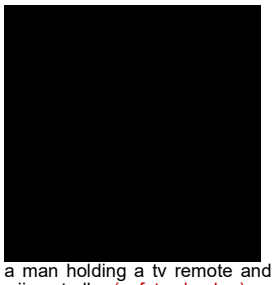
Sample testings

Data and embedding checks implemented in `notebooks/tests.ipynb`: subset has 2,000 rows; no missing images or empty captions; all 80 classes present; every image is 512x512; each embedding has shape 77 x 768 with no NaNs; related captions score higher cosine similarity than unrelated ones.

Baseline generation (CPU, fp32, 25 steps, about 22 s per step, 9.3 to 11.2 min per image):



a cellphone, piece of fruit and cup are on the table



a man holding a tv remote and wii controller (safety checker)



the fruits are all in a container in the window



a large bird perched on top of a stick near a window



fire hydrant on city street with several parked cars

Observations. Scene layout and style are strong: the window, fruit, bird and street scenes all match their captions. Weaknesses show up in object completeness and counting: the table scene drops the cup, and the hydrant caption (one hydrant) produced three. The bird has minor anatomical artifacts. One benign caption was blacked out by the built-in safety checker, a false positive we must handle when generating at scale. CPU generation is far too slow for full-subset evaluation, so later milestones will need GPU access.